

InstBlink++: Real-Time Multi-Person Eyeblink Detection in Untrimmed In-the-Wild Videos

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Abstract—Real-time eyeblink detection in unconstrained scenarios supports diverse applications like fatigue monitoring and emotion analysis. Existing methods are typically designed under a single-person, trimmed-video assumption, overlooking the multi-instance and untrimmed nature of real-world scenarios, which brings inherent challenges in achieving consistent multi-instance awareness and fine-grained instance-aware eyeblink boundary localization. In this work, we address this gap through comprehensive contributions covering the dataset, methodology, and evaluation aspects. In particular, we construct MPEblink, a high-quality dataset for unconstrained, multi-instance-aware eyeblink detection in untrimmed videos, which also serves as the largest eyeblink detection corpus to date. Interpretable metrics are also designed for robust multi-dimensional model evaluation. On the methodological side, we propose InstBlink, the first one-stage, instance-aware multi-person eyeblink detection framework with end-to-end learning capability. InstBlink jointly addresses face detection, face tracking, and instance-level blink detection, offering two key benefits: (1) more effective utilization of shared face-eye features, and (2) constant-time ($O(1)$) complexity with respect to the number of instances, enabling highly efficient inference. Additionally, in our new version, InstBlink++, we introduce disentangled fine-grained eye modeling and global-local feature interaction, achieving substantial performance gains while preserving the real-time advantages. Our experiments uncover both the intrinsic difficulty of real-time multi-person eyeblink detection in untrimmed in-the-wild videos and the surprising weakness of state-of-the-art large vision-language models on this fine-grained task. InstBlink surpasses existing approaches by large margins while maintaining high inference speed. The source code and dataset are available at <https://github.com/wenzhengzeng/MPEblink>.

Index Terms—Multi-person eyeblink detection, eyeblink in the wild, untrimmed video, one stage.

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I. INTRODUCTION

REAL-TIME eyeblink detection in unconstrained scenarios has recently emerged as a challenging research task [1], with applications including fatigue detection [2], face anti-spoofing [3], and affective analysis [4]. Although remarkable progress has been made [1], [5], [6], existing works predominantly focus on single-person assumption in trimmed settings, leaving multi-person eyeblink detection in untrimmed videos largely underexplored.

However, in many real-world settings, practical scenarios involve an arbitrary number of interacting individuals over naturally untrimmed temporal spans. In such settings, eyeblink dynamics provide fine-grained and objective non-verbal behavioral cues that have been widely studied in psychology and affective computing and shown to correlate with cognitive load, fatigue, and engagement [4], [7]–[11]. Extending eyeblink analysis to unconstrained scenarios with an open-ended number of individuals enables scalable, instance-level modeling of behavioral dynamics in realistic social environments (e.g., classrooms, meetings, and outdoor group activities), where simultaneous monitoring of multiple participants across unbounded time is required. Such requirements are difficult to fulfill with single-person and trimmed-video-focused approaches, thus motivating the development of an effective and efficient eyeblink detection system that supports fully unconstrained environments with varying numbers of individuals and untrimmed temporal contexts.

To this end, we shed the light on this research problem with essential contributions on dataset, methodology, and evaluation. First, a challenging, high-quality, human-labeled multi-person eyeblink detection dataset termed MPEblink is built, supported by a clearly defined problem setting. It consists of untrimmed videos captured from diverse unconstrained movies and television programs to reveal realistic in-the-wild characteristics. The scenarios are diverse, including social interactions and group activities, and pose substantial challenges for effective eyeblink detection, as verified in our later analysis. Fig. 1 (a) illustrates a sample video with ground truth annotations. Unlike existing eyeblink detection benchmarks [1], [8], [12], the proposed setting explicitly requires models to remain aware of all attendees throughout the video and detect their eyeblink events with precise temporal boundaries at the instance level, offering a closer correspondence to real-world demands. Along with this, new metrics are proposed to address the lack of a unified and fair evaluation emphasizing both instance awareness and instance-level eyeblink boundary

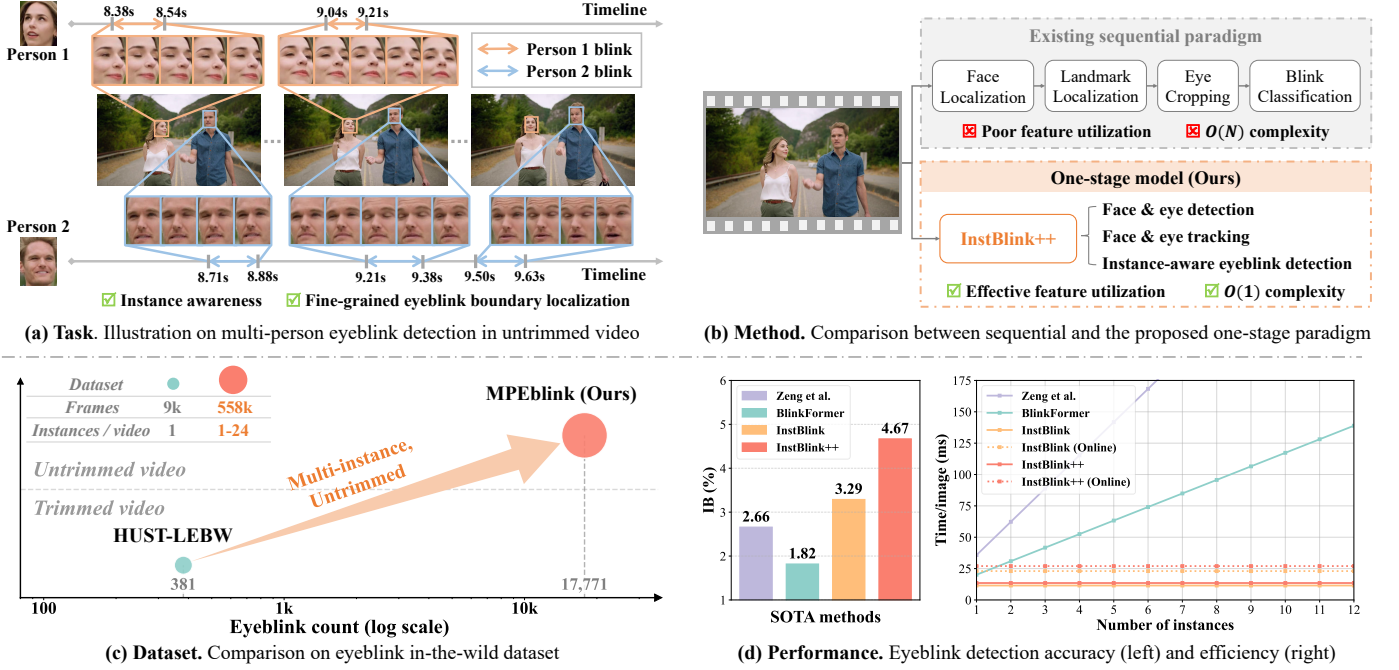


Fig. 1. (a) **Task:** The illustration on multi-person eyblink detection in untrimmed video. This task aims at being aware of all the people present and detecting their eyblink activities at instance level. (b) **Method:** Comparison between traditional sequential approaches and the proposed one-stage paradigm. (c) **Dataset:** We extend the in-the-wild concept to more general multi-instance and untrimmed scenarios, with meticulously designed interpretable metrics for robust evaluation. (d) **Performance:** Our method demonstrates clear advantages in both accuracy and efficiency. More details in Sec. VI.

detection quality. In summary, MPEblink is featured with multi-instance, unconstrained, and untrimmed, which makes it more challenging and realistic than previous formulations. Our upgraded version—improved in label quality, scenario diversity, and evaluation metrics—serves as the most comprehensive eyblink dataset to date for both model development and assessment, as shown in Fig. 1 (c).

To perform eyblink detection, previous methods [1], [6], [8], [12], [13] typically follow a sequential process involving the sub-tasks of face detection, face tracking, and classification on the pre-extracted local eye clues within a temporal window (Fig. 1 (b)). Although this pipeline appears reasonable, it possesses several notable limitations. Firstly, the sub-tasks run in an isolated way, lacking joint optimization and interaction to ensure global optimal performance. As known, they are actually highly correlated to each other. Besides, it is also inefficient due to the inability to reuse the features of each stage. Secondly, the eyblink features are solely derived from the pre-extracted local eye clues (i.e., a small part of face), lacking useful information from global face context such as head pose and illumination conditions that are crucial for detecting eyblinks reliably in the wild. Moreover, the reliability of the pre-extracted local eye clues also cannot be guaranteed, due to the challenges of accurately localizing them in unconstrained scenarios. Thirdly, the sequential approach leads to the computational cost dependent on the subject amount, which makes it hard to meet the real-time running requirement.

To address this, we first propose InstBlink, a one-stage framework for multi-person eyblink detection. Different from existing methods, it simultaneously addresses multiple tasks:

detecting human faces, tracking them over time, and localizing their eyblink behaviors at instance level (Fig. 1 (b)). InstBlink takes inspiration from the existing query-based methods [14]–[17] and models the spatio-temporal instance-level face as well as eyblink representations within each query (we denote it as instance query). The insight is that the features can be effectively shared among these sub-tasks and the eyblink features can be facilitated via face’s global contexts (e.g., head pose and illumination condition) with joint optimization and interaction, especially in unconstrained in-the-wild cases. Benefiting from our one-stage design, Instblink demonstrates high efficiency and the computational cost is insensitive to the subject amount in the scene.

Despite its effectiveness, we observe that the shared query design in InstBlink leads to feature entanglement and insufficient modeling of fine-grained eye-region cues—both of which are critical for precise blink discrimination. To address this, we introduce an upgraded version, InstBlink++, where each instance query is equipped with paired eye queries to explicitly capture fine-grained eye-specific features and enhance discriminative power. This design is further complemented by a decoupled decoding strategy with an eye-focus constraint. We also introduce a global-local feature interaction mechanism to enable the global awareness of the eye modeling. InstBlink++ delivers substantially improved performance over the initial InstBlink, achieving relative gains of 41.9% and 33.6% on IB and Blink-AP, respectively, while retaining the real-time efficiency afforded by the one-stage design.

Our extensive experiments show that existing methods—including recent large video-language models [18] with strong general-purpose performance—perform poorly on the

MPEblink dataset, highlighting the critical challenges of unconstrained multi-person eyeblink detection in untrimmed videos. Our InstBlink series significantly outperform existing arts by large margins with a high inference speed (as shown in Fig. 1 (d)). The main contributions of this work are summarized as follows.

- We take the pioneering step in defining and exploring multi-person eyeblink detection in untrimmed videos, paving the way for future research.
- We construct MPEblink, the first unconstrained multi-person eyeblink detection dataset, together with interpretable metrics for multi-dimensional evaluation. Our upgraded version serves as the most comprehensive dataset to date for model development and assessment.
- We propose InstBlink, the first one-stage, instance-aware multi-person eyeblink detection framework, delivering notable gains in both accuracy and efficiency. Our upgraded version, InstBlink++, further strengthens fine-grained blink modeling through disentangled eye modeling, achieving significantly better performance while preserving real-time efficiency.

The preliminary version of our work appeared in [19]. Compared with it, we introduce substantial improvements and extensions in three aspects: (1) **Dataset upgrade.** We upgrade MPEblink 1.0 to MPEblink 2.0 by providing higher-quality annotations, expanding scenario diversity with a particular emphasis on dense (up to 20+ co-exist instances) and long-video settings, and introducing an enhanced evaluation metric, IB, as a unified metric that mitigates the bias of the two standalone metrics used in our preliminary version [19]. (2) **Methodological enhancement.** We identify a feature-entanglement issue in InstBlink and address it by introducing paired eye queries and a decoupled decoding strategy for finer-grained blink modeling. A global-local feature interaction mechanism is also introduced to strengthen global context awareness. InstBlink++ achieves significantly better performance than the initial InstBlink while preserving the real-time efficiency of the one-stage design. (3) **Extended analysis.** We provide a more in-depth description and experimental analysis. We uncover the inability of general action detectors and even recent popular large vision-language models to handle such fine-grained blink detection. We further validate the superiority and generalization capability of our proposed method.

II. RELATED WORK

A. Eyeblink Detection Dataset

Early datasets [3], [8], [12], [20], [21] primarily focus on constrained indoor cases where volunteers sit directly in front of the camera under consistent environmental conditions. Later efforts extend toward less constrained scenarios: mEBAL2 [22] expands the mEBAL dataset [8] with additional variations in facial attributes, lighting, pose, and camera distance, yet the overall diversity remains limited due to the controlled laboratory environment. Fogelton and Benesova [23] introduce the Researcher’s Night dataset where volunteers are often acting naturally (e.g., touching their face, moving their head or even talking to somebody) with cluttered

backgrounds. Nevertheless, the variation is still limited in aspects of illumination, camera distance, etc [1]. Another common limitation remains that volunteers are aware of being recorded, which can lead to unnatural eyeblink behaviors [23].

HUST-LEBW [1] is a pioneering attempt that pushes the eyeblink detection research field towards unconstrained scenarios. Specifically, factors such as face attributes, pose, and illumination are highly diverse and unrestricted. Moreover, unlike the volunteers in existing datasets (e.g., [23]), actors in the movies are not informed about the recording of their eyeblink behavior, resulting in more natural eyeblink behaviors. However, HUST-LEBW mainly focuses on single-person scenarios, which limits the application of methods in more realistic scenarios such as group activities and social interactions. Another issue is that it mainly focuses on eyeblink recognition task in trimmed short videos and only provides very limited untrimmed samples (i.e., 90) for testing purposes. As known, for real-world demands, detecting/localizing eyeblink in untrimmed videos is actually more preferred than mere recognition in trimmed ones.

To tackle these limitations, we introduce MPEblink, the first dataset to formalize multi-instance eyeblink detection in untrimmed in-the-wild videos, along with metrics that evaluate method capability across different aspects, laying a solid foundation for future research.

B. Eyeblink Detection Method

Existing methods generally judge eyeblinks from the pre-extracted local eye clues (e.g., local eye region [1], [5], [6], [8], [24]–[29] or landmarks around eyes [13], [30]–[32]). To obtain the local eye clues, the existing works generally run in a sequential way including face detection, face tracking, and landmark detection. Finally, the eyeblink result is classified from the pre-extracted local eye clues within a temporal window. Such a pipeline is inefficient due to the isolated optimization among different sub-tasks. Meanwhile, the eyeblink features only contain the information from the pre-extracted local eye clues, lacking useful global information such as head pose and illumination, and the pre-extracted eye clues are also unreliable due to the localization challenge toward unconstrained cases. Besides, due to the multi-stage characteristic, existing methods do not scale well with the number of people in the scene to meet the real-time running requirement. In this paper, we propose the first one-stage, instance-aware multi-person eyeblink detection framework with joint multi-task learning and execution capability, achieving substantial improvements in both performance and efficiency.

C. Spatio-temporal Action Detection

Spatio-temporal action detection [15], [33]–[42] aims at simultaneously obtaining the spatial and temporal location of an action, which is defined as an action tube [37]. The task mainly focuses on the isolated coarse-grained action tubes (e.g., jump, walk) and lacks instance awareness, thus failing to support instance-level analysis throughout the video. Different from such task formulation, our proposed multi-person eyeblink detection task aims at simultaneously being



Fig. 2. The snapshots within the MPEblink dataset. The proposed dataset is featured with more realistic and diverse scenarios.

aware of the presence of people and analyzing their fine-grained eyeblink behaviors at the instance level.

D. Query-based Method

Query-based methods are impacting many computer vision tasks. After DETR [43], the pioneer that introduces the query-based framework to address object detection, numerous efforts have been subsequently paid to facilitate visual recognition under the query concept in the fields of object detection [14], [44]–[50], pose estimation [51], [52], gaze estimation [53]–[56], action detection [15], [57], and segmentation [16], [17], [58]–[62]. Inspired by these developments, we extend the query-based paradigm to multi-instance eyeblink detection, facing essential challenges in consistent instance tracking and fine-grained eyeblink detection that prior works have largely underexplored and left unresolved. We predict a set of identity-aware queries with a pairing design, which jointly encode a two-level hierarchical representation, capturing spatio-temporal identity information at the face level and fine-grained eye dynamics of eyeblinks. This joint hierarchical design enables accurate and efficient modeling of instance-aware eyeblink behaviors in unconstrained and untrimmed videos.

III. THE MPEBLINK DATASET

Existing eyeblink detection datasets generally focus on single-person cases and are often restricted to constrained settings or trimmed videos [1], [3], [8], [12], [20], [22], [23], [28]. To explore unconstrained eyeblink detection under multi-person and untrimmed scenarios, we construct a large-scale multi-person eyeblink detection dataset termed MPEblink to shed the light on this research topic that has not been well studied before. The distinguishing characteristics of MPEblink lie in 3 aspects: multi-person, unconstrained, and untrimmed long video, which makes it more realistic and challenging than previous datasets [1], [3], [8], [12], [20], [22], [23], [28].

A. Task Formulation

We think that a good multi-person eyeblink detection algorithm should be able to (1) detect and track human instances’

faces reliably to ensure the instance-level analysis ability along the whole video, and (2) detect temporal eyeblink boundaries accurately within each human instance to ensure the precise awareness of their eyeblink behaviors.

Formally, for an untrimmed video with T frames and N_{gt} people in it, towards the j -th person, let $\hat{L}^j \in \mathbb{R}^{T \times 4}$ denote its face bounding boxes across the video. $\hat{C}^j \in \mathbb{R}^T$ is given as the face labels to reflect the face existence of this instance in each frame (i.e., $\hat{C}_t^j \in \{1, 0\}$, where $\hat{C}_t^j = 0$ indicates that the face is not visible due to serious occlusion or departure at time t). $\hat{L}_t^j$ is also set to $\emptyset$ when the person is not visible. We use $\hat{L}^j$ and $\hat{C}^j$ to represent the instance-level information of each person. Suppose person j blinks $\hat{K}^j$ times in the video. Let $\hat{B}_k^j = [\hat{s}_k^j, \hat{e}_k^j]$ denote the $\hat{k}$ -th eyeblink event within this person, where $\hat{s}_k^j$ and $\hat{e}_k^j$ denote its starting and ending time. Suppose a multi-person eyeblink detection algorithm produces H human instance hypotheses. Within each human instance hypothesis i , it needs to produce a sequence of predicted face bounding boxes L^i , face classification scores C^i , and K^i eyeblink proposals $B_k^i = [s_k^i, e_k^i]$. Better performance can be acquired when predictions get closer to ground truths.

B. Data Collection

We collect video clips from diverse unconstrained movies and television programs, yielding samples that exhibit rich “in-the-wild” characteristics across a wide range of indoor and outdoor scenarios (Fig. 2). While such data are more realistic and closer to practical use cases, they also introduce substantial challenges for multi-person eyeblink detection. For instance, the pose, illumination condition, and face size among instances differ a lot, even within the same video, as shown in Fig. 3. Moreover, because of the various types of interaction among instances, there may exist severe occlusion or people in and out scenarios as shown in Fig. 4, making it harder for both instance localization and their eyeblink event detection. Besides, different from the existing unconstrained counterpart [1] that mainly focuses on trimmed videos, MPEblink targets on untrimmed setting. As a result, we not only need to recognize eyeblink but also need to accurately localize the starting and ending time of each eyeblink event, which is more challenging especially



Fig. 3. Essential challenges caused by variation on pose, scale, and illumination in the MPEblink dataset.



Fig. 4. Occlusion and people in & out cases during interaction among instances.

when rapid and consecutive eyeblinks occur. An example is shown in Fig. 5. It can be observed that the appearance difference among eye statuses is small when consecutive rapid eyeblinks happen, making it hard to distinguish the boundary of each eyeblink. Considering the above challenges together, it can be summarized that unconstrained multi-person eyeblink detection in untrimmed videos is indeed a challenging research task, which is further validated by our experiments in Sec. VI.

C. Data Annotation

The main data annotation work on MPEblink is shown in Fig. 6. For human instances in each video, we labeled their face bounding boxes exhaustively across the whole video. To increase the dataset’s versatility for a broader range of applications, 68 facial landmark positions [63] are also annotated. Technically, this part of the annotation is under a semi-supervised manner: we use the state-of-the-art face analysis engine InsightFace [64] to achieve face bounding box [65] and landmark [63] detection (note that at the same time, the eye region bounding box can also be acquired by setting a box based on the facial landmarks around eyes). To track instances across frames, a matching strategy that considers face bounding box IoU and similarity among deep face features [66] is designed. Besides, human annotators carefully check the annotation quality and correct the wrong bounding boxes and tracking results generated by the algorithm. Note that if an human instance disappears because of full occlusions and then re-appears, the instance identity should keep the same.

For eyeblink events within each instance, human annotators carefully define their start and end frame indices. To ensure annotation quality and consistency, we establish a detailed and unified annotation protocol grounded in the standard definition of eyeblink: the complete process of eyelid closure and reopening [67], [68]. Specifically, we define the start index as the frame when the degree of eye opening first begins to decrease, and the end index as the frame when the eyes reopen, and the degree of opening remains stable without significant variation or returns to the normal state. Annotators are trained

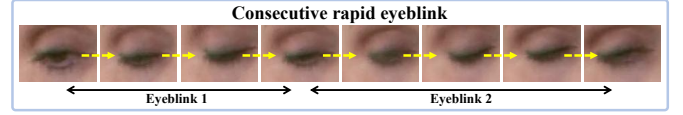


Fig. 5. An example of rapid and consecutive eyeblinks in untrimmed video.

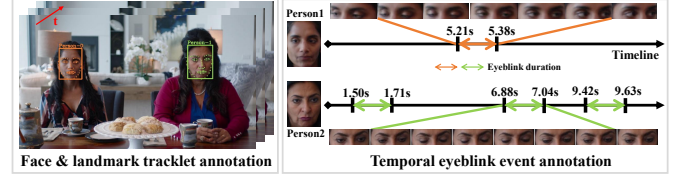


Fig. 6. The main annotation works of the MPEblink dataset.

through multiple rounds until their annotations conform to the established protocol. Annotating eyeblinks is highly time-consuming: on average, labeling a single human instance over one minute of video requires approximately 15 minutes of manual effort.

D. The MPEblink Series

Our preliminary version, MPEblink 1.0 [19], contains 686 untrimmed videos and 8,748 eyeblink events. This version already serves as a comprehensive benchmark for developing and evaluating unconstrained multi-person eyeblink detection in untrimmed videos.

Nevertheless, MPEblink 1.0 still exhibits several limitations. (1) It primarily focuses on relatively sparse scenarios, where videos typically contain only one to three individuals and rarely more than eight, which limits its ability to reveal the challenges inherent in dense crowd scenes. (2) Although it targets untrimmed eyeblink videos, the video durations are relatively short—most ranging from 10 to 20 seconds—thus failing to fully capture the complex dynamics of longer videos with richer variations in individuals, scenes, and interactions that are challenging for robust eyeblink detection. (3) We observe that MPEblink 1.0 still contains some inaccurate annotations, involving both instance localization and the temporal boundaries of eyeblink events.

To address these limitations, we upgrade the dataset to MPEblink 2.0, with the following core improvements. (1) We review and refine the annotation quality of MPEblink 1.0. Specifically, we conduct a comprehensive round of frame-level annotation verification and refinement across all videos in MPEblink 1.0 to ensure higher ground-truth accuracy and annotation consistency. (2) We further include 235 new video samples that deliberately emphasize (i) dense crowd scenes, with up to more than 20 concurrent instances, to reveal challenges in dense multi-person eyeblink detection under realistic application scenarios (e.g., classroom, meeting, and outdoor group activity, as shown in Fig. 7), and (ii) longer video, extending up to 300 seconds, to better expose and investigate inherent complex temporal dynamics. We argue that minute-level videos already provide a meaningful benchmark for studying dynamic temporal behaviors in video, while hour-level videos—whose difficulty is often dominated by long-term instance identity tracking rather than eyeblink understanding—are left for future exploration. This design choice

TABLE I
HIGH-LEVEL STATISTICS OF MPEBLINK AND EXISTING EYEBLINK DETECTION DATASETS. s: SECONDS, m: MINUTES, f: FRAMES.

Dataset	Num. of videos	Video length	Unconstrained?	Instances/video	Eyeblink count
Talking Face [21]	1	3.3m	×	1	61
ZJU [3]	80	4.1-4.7s	×	1	255
Eyeblick8 [12]	8	2.7-8.9m	×	1	353
Silesian5 [20]	5	1.4-2.7m	×	1	300
Researcher's Night [23]	107	0.3-3.4m	×	1	1,867
mEBAL [8]	6,000	19f	×	1	3,000
mEBAL2 [22]	21,100 / 180	19f / 15-40m	×	1	10,550
HUST-LEBW [5]	673 / 90	13f / 0.7-49.4s	✓	1	381 / 176
MPEblink 1.0	686	7.1-85.9s	✓	1-8	8,748
MPEblink 2.0	921	4.1-300s	✓	1-24	17,711



Fig. 7. Samples with different scenario and instance layout within MPEblink 1.0 and MPEblink 2.0.

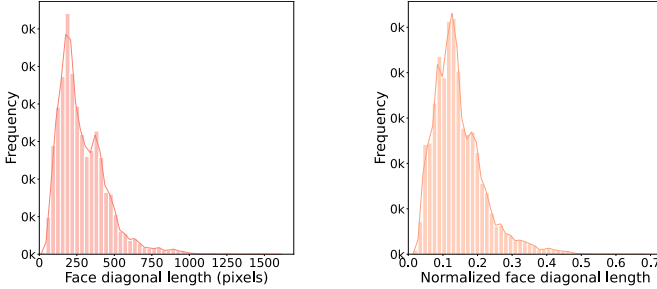


Fig. 8. (a) The box size of face. (b) The normalized box size of face.

makes the dataset cover a broader range of scenarios while remaining focused on the core task.

E. Data Statistics

We analyze the statistics of MPEblink 2.0. In Fig. 8, we show the distributions of face sizes in both (a) absolute terms and (b) relative to the image size. The absolute size is computed as the diagonal length (in pixels) of face bounding boxes, while the relative size is obtained by normalizing this value by the image diagonal. A striking variation is observed in both absolute and relative scales, reflecting the diverse and unpredictable nature of real-world scenarios. From Fig. 9, we further observe that (1) instances (via face center localization) appear across a wide range of spatial positions, and (2) the distances between instances within the same scene vary substantially, indicating complex inter-instance spatial relations. In Fig. 10, we also plot the distributions of head pose and gaze direction (estimated using [69] and [55]), which similarly exhibit high diversity.

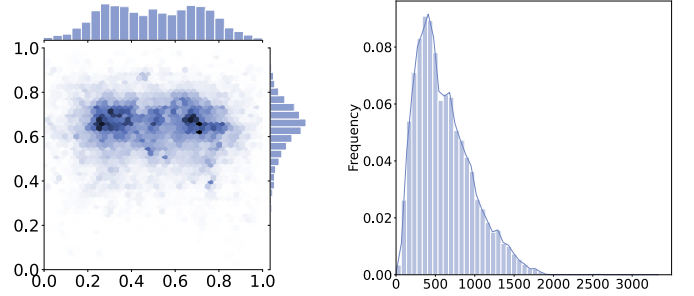


Fig. 9. (a) Distribution of the face location over the whole image. (b) The distance (pixels) among instances within the same images.

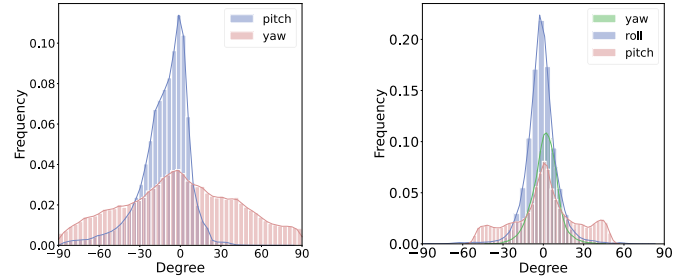


Fig. 10. Distribution of (a) eye gaze and (b) head pose within MPEblink.

Fig. 11 visualizes the duration distributions of videos and blink events. As shown, blink events are extremely short relative to untrimmed videos (89.42% last under 0.4 s and 99.22% under 0.8 s) and are markedly shorter than general actions (e.g., high jump, cliff diving), which typically span several seconds with larger spatial motion [70]–[72]. Such brevity makes precise blink-boundary localization highly challenging and demands fine-grained spatiotemporal modeling to capture their rapid and subtle dynamics.

Tab. I compares the MPEblink series with existing eyeblink detection datasets. MPEblink 2.0 contains the largest number of eyeblink events (17,771 across 921 untrimmed videos, double the size of v1.0 in our conference version [19]). Although SynBlink [28] includes more events, it is synthetic and exhibits a significant domain gap when applied to real-world data [28]. A key distinction of MPEblink is its wide range of human instances per video (1–24) with exhaustive annotations, captured under fully unconstrained in-the-wild conditions, as illustrated in Fig. 2. Moreover, existing in-the-wild dataset [1] mainly focuses on trimmed cases and only provides a limited number (i.e., 90) of untrimmed videos for testing purposes.

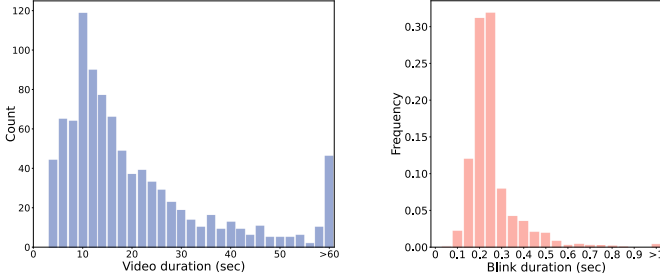


Fig. 11. Distribution of video duration (left) and blink duration (right).

In contrast, MPEblink is specifically designed for untrimmed scenarios, supporting both model training and evaluation, and offers substantially greater quantity, diversity, and more fine-grained annotations, making it markedly more practical for real-world deployment and research benchmarking.

F. Evaluation Metrics

Existing metrics for eyeblink detection only focus on single-person cases without multi-instance awareness. To address this limitation we introduce 3 new metrics termed Inst-AP, Blink-AP, and IB to give attention to both instance awareness quality and eyeblink detection quality.

Inst-AP. It aims to evaluate the instance detection and tracking ability of methods, which is the basis for analyzing the eyeblink behaviors of each instance. We modify the standard evaluation metric video-AP in spatio-temporal action detection [37] to fit our task. Different from the original implementation, the 3D IoU is calculated between each instance proposal (i.e., a sequence of face bounding boxes) and ground truth across the whole video, rather than the action tube only. As a result, the proposed metric can reveal the algorithm’s ability for detection and tracking instances, while video-AP can just reveal the localization ability of the isolated action tube without instance-level awareness. We report the averaged Inst-AP over IoU thresholds from 50% to 95% with a step size of 5%, together with Inst-AP₅₀ and Inst-AP₇₅ evaluated at individual IoU thresholds. During calculating Inst-AP, each obtained true positive (TP) prediction has matched with one ground-truth instance, and thus, we can calculate the eyeblink detection accuracy of these TP predictions using the proposed Blink-AP. The TP predictions under IoU of 50% are used to calculate the subsequent Blink-AP.

Blink-AP. This metric is used to reflect the model’s eyeblink detection ability within each instance. We tailor the standard AP metric in temporal action detection [71] into our task. Only the matched (i.e., true positive) predictions when calculating Inst-AP are taken into consideration in this metric, as the blink accuracy of these predictions is rarely affected by the face detection and tracking accuracy. We report the averaged Blink-AP over IoU thresholds from 50% to 95% with a step size of 5%, together with Blink-AP₅₀ and Blink-AP₇₅ evaluated at individual IoU thresholds.

IB. Despite their usefulness, the above metrics may introduce biases when assessed independently. For example, a method may obtain a low Inst-AP if it detects only a small

subset of easy instances, yet still reports an inflated Blink-AP because those instances also contain easier blink cases. Conversely, a method with high Inst-AP—by localizing more instances—may show a lower Blink-AP as harder instances and challenging blink events are included. These imbalances motivate a unified metric that jointly reflects both instance localization and blink detection quality.

To address this, in this extension, we further design a unified metric IB. Inspired by the IMQ metric [73], IB is defined as:

$$IB = \text{Blink-AP} \cdot \frac{|TP|}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|}, \quad (1)$$

where $|TP|$, $|FP|$, and $|FN|$ denote the numbers of true positive, false positive, and false negative instance predictions, respectively. The second term is an F_1 -like score that measures instance localization quality, and serves to modulate the blink-detection performance measured by Blink-AP. Overall, IB captures both instance localization and blink-detection quality, providing a more balanced and reliable evaluation that mitigates the biases of the two standalone metrics. We report IB, computed using Blink-AP (in Eq. 1) averaged over IoU thresholds from 50% to 95% with a step size of 5%, as well as IB₅₀ and IB₇₅, computed using Blink-AP at IoU thresholds of 50% and 75%, respectively.

IV. INSTBLINK

In this section, we present our preliminary version InstBlink that takes a video clip as input and directly outputs the face positions of each instance and their eyeblink intervals in the whole video clip. After that, we will introduce the advanced version, InstBlink++, in Sec. V.

The overall architecture of Instblink is illustrated in Fig. 12. Given a video clip $I \in \mathbb{R}^{T \times 3 \times H \times W}$ where T denotes the number of frames and $H \times W$ is the spatial size of each frame, InstBlink first applies a backbone network to extract video feature $F_v \in \mathbb{R}^{T \times D \times H' \times W'}$ where D is the channel number and $H' \times W'$ is the spatial size of the feature. Afterward, a query-based architecture iterates M times for gradual instance-level feature extraction and representation, which consists of 3 components: the Query Interaction Module (QIM), Video Interaction Module (VIM), and task-specific heads (i.e., face head for face detection and tracking, and blink head for instance-level eyeblink detection). At the end of each interaction, the queries will be updated and the instance-level face and eyeblink predictions will be output by the task-specific heads. During inference, the output from the last iteration will be used as the final prediction results. Next, InstBlink will be illustrated in more detail.

A. Instance Query

We build spatio-temporal instance queries $\{Q^i\}_{i=1}^N$ to jointly capture each human instance’s facial features and eyeblink dynamics throughout the video. Each query contains T embeddings (i.e., $Q^i \in \mathbb{R}^{T \times D}$), where T denotes the number of video frames (one embedding per frame) and D is the feature dimension. Each embedding generally focuses on the instance’s face and eyeblink representations in the corresponding frame.

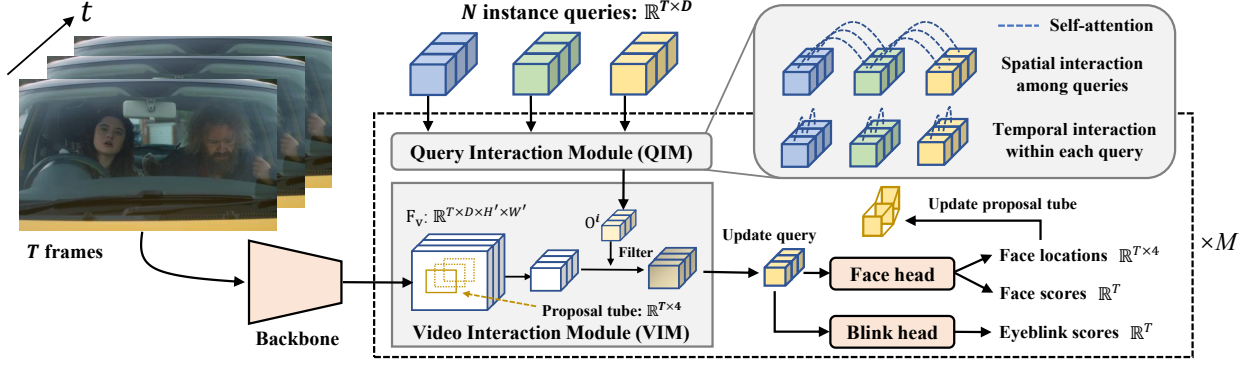


Fig. 12. Overview of the InstBlink framework.

Each query is also paired with a proposal tube $P^i \in \mathbb{R}^{T \times 4}$ (i.e., spatio-temporal bounding boxes across time), which aims at indicating the face location of i -th instance across the entire video clip. At the first iteration of each complete forward propagation, Q^i and P^i are initialized by copying 2 learnable parameters $\tilde{Q}^i \in \mathbb{R}^{1 \times D}$ and $\tilde{P}^i \in \mathbb{R}^{1 \times 4}$ T times along the temporal dimension.

B. Query Interaction Module (QIM)

QIM targets at (1) enhancing the association between the specific query and its corresponding human instance and (2) modeling spatio-temporal face and eyeblink representations of the associated instance. Specifically, we first adopt a spatial self-attention layer to allow spatial interaction among instance queries within each frame:

$$\{Q_t^i\}_{i=1}^N = \text{MHSA} \left(\{Q_t^i\}_{i=1}^N \right), \quad t \in [1, T], \quad (2)$$

where MHSA is the multi-head self-attention [74]. Spatial interaction can build strong communication among instance queries to better model the instance-level features under complex circumstances such as occlusion due to human interactions. Then, temporal self-attention is used within instance query along the temporal dimension to realize temporal interaction:

$$\{Q_t^i\}_{t=1}^T = \text{MHSA} \left(\{Q_t^i\}_{t=1}^T \right), \quad i \in [1, N]. \quad (3)$$

Applying temporal interaction within each query allows the embeddings from different frames to communicate with each other to facilitate instance tracking and model temporal eyeblink representations of the corresponding instance.

C. Video Interaction Module (VIM)

VIM aims at collecting the face and eyeblink information for each instance from the video feature. Particularly, dynamic filters [14] O^i are first generated from each query embedding Q^i , and are applied via dynamic convolution to an RoI feature to capture highly relevant face-eye cues. The RoI feature is obtained by applying RoIAlign [75] on the video feature using the proposal tube P^i . The filtered feature is then linearly projected with a residual connection to produce the updated query feature $\tilde{Q}^i$, which is subsequently used for both face and eyeblink predictions.

Frame index t	1	2	3	4	5	6	7	8	9	10	11	12
Frame-level eyeblink scores b^t	0.2	0.1	0.5	0.6	0.6	0.4	0.2	0.4	0.7	0.6	0.1	0.1
$1_{b_t^i > \text{threshold}(0.3)}$	0	0	1	1	1	1	0	1	1	1	0	0
Eyeblink interval B^i												

Fig. 13. An example of the blink merging procedure, which takes the frame-level eyeblink scores b^i and outputs the interval-level eyeblink results B^i . The threshold is set to 0.3 for demonstration.

D. Task-specific Heads

The instance-level face and eyeblink predictions can be obtained by applying task-specific heads on the query features.

Face head. Given the updated instance query feature $\tilde{Q}^i$, a Multilayer Perceptron (MLP) layer with Sigmoid normalization is used to indicate the existence of human faces as

$$C^i = \text{Sigmoid} \left(\text{MLP}_C \left(\tilde{Q}^i \right) \right), \quad (4)$$

where $C^i \in \mathbb{R}^T$ denotes the face classification scores across frames. Face localization is achieved in a similar way as

$$L^i = \text{MLP}_L \left(\tilde{Q}^i \right), \quad (5)$$

where $L^i \in \mathbb{R}^{T \times 4}$ denotes the face bounding boxes across frames. It will also be used to renew the proposal tube P^i .

Blink head. Eyeblink prediction is obtained considering both instance and eye query features:

$$b^i = \text{Sigmoid} \left(\text{MLP}_b \left(\tilde{Q}^i \right) \right), \quad (6)$$

where $b^i \in \mathbb{R}^T$ is the frame-level eyeblink scores with each element b_t^i indicating the likelihood that the t -th frame lies within an eyeblink event. During inference, we simply merge the adjacent frame-level eyeblink predictions based on a threshold (e.g., 0.3 in our implementation) to form interval-level predictions $B_k^i = [s_k^i, e_k^i]$ where k indexes the predicted eyeblink intervals. An intuitive example of the merging procedure is shown in Fig. 13.

Unlike existing methods that rely solely on local eye-region features, our blink head also considers useful global information such as face appearance, head pose, and illumination condition to aid implicit eye localization and robust blink characterization. This is because the blink head directly filters global features encoded in the query embeddings, where rich

face context is already stored. During training, the eyeblink clues will not only flow back to the RoI feature, but also to the instance query to enhance the eyeblink-related feature representation ability of queries. Moreover, because of the shared face-eye feature, a kind of interaction and synergy is built where the information flow from blink head can also benefit the face representations to boost the face localization capability, as verified in Sec. VI-F.

As we have N instance queries, we finally obtain the predictions from them in parallel:

$$\{y^i\}_{i=1}^N = \{(C^i, L^i, b^i)\}_{i=1}^N. \quad (7)$$

E. Training

To optimize InstBlink, one-to-one assignment between instance-level predictions $\{y^i\}_{i=1}^N$ and ground truths is first conducted based on their instance-level face tracklet similarity. Then, the loss is calculated concerning face and eyeblink items jointly for optimization. As in Sec. III-A and Sec. III-C, the ground truth annotations can be summarized as

$$\{\hat{y}^j\}_{j=1}^{N_{gt}} = \left\{ \left(\hat{C}^j, \hat{L}^j, \hat{b}^j \right) \right\}_{j=1}^{N_{gt}}, \quad (8)$$

where $\hat{C}^j \in \mathbb{R}^T$, $\hat{L}^j \in \mathbb{R}^{T \times 4}$ denotes the frame-level face existence and face position (i.e., bounding box), respectively. $\hat{b}^j \in \mathbb{R}^T$ indicates eyeblink existence (i.e., $\hat{b}_t^j = 1$ indicates frame t is inside an eyeblink event and conversely for $\hat{b}_t^j = 0$, which can be derived from $\hat{B}^j$ using an inverse process in Fig. 13). We use Hungarian algorithm [76] to perform instance-level bipartite matching between predictions and ground truths. The matching cost is given as

$$\mathcal{L}_{\text{Hung}}(y^i, \hat{y}^j) = \sum_{t=1}^T \left[\mathcal{L}_{cls} \left(C_t^i, \hat{C}_t^j \right) + 1_{\{\hat{C}_t^j \neq 0\}} \left(\mathcal{L}_{\text{box}} \left(L_t^i, \hat{L}_t^j \right) \right) \right], \quad (9)$$

where $\mathcal{L}_{cls}$ indicates the focal loss [77] for face existence classification. $\mathcal{L}_{\text{box}}$ is the combination of L1 loss and GIoU loss [78] for face localization. The loss weights follow the common setting within object detection fields [14], [16]. After obtaining the optimal assignment $\hat{\sigma}$ between predictions and ground truths, the network is optimized by the loss as

$$\mathcal{L} = \sum_{j=1}^{N_{gt}} \left(\mathcal{L}_{face} \left(\hat{y}^j, y^{\hat{\sigma}(j)} \right) + \lambda \mathcal{L}_{blink} \left(\hat{y}^j, y^{\hat{\sigma}(j)} \right) \right), \quad (10)$$

where $y^{\hat{\sigma}(j)}$ is the prediction that has been matched with the ground truth instance $\hat{y}^j$. $\mathcal{L}_{face}$ is the face tracklet detection loss that shares the same form as $\mathcal{L}_{Hung}$ (i.e., $\mathcal{L}_{face}(\hat{y}^j, y^{\hat{\sigma}(j)}) = \mathcal{L}_{Hung}(\hat{y}^j, y^{\hat{\sigma}(j)})$), and $\mathcal{L}_{blink}$ denotes instance-level eyeblink detection loss as $\mathcal{L}_{blink}(\hat{y}^j, y^{\hat{\sigma}(j)}) = \sum_{t=1}^T \mathcal{L}_{cls}(\hat{b}_t^j, b_t^{\hat{\sigma}(j)})$ with $\lambda = 5$. For unmatched predictions, only $\sum_{t=1}^T \mathcal{L}_{cls}(C_t^i, \hat{C}_t^i = 0)$ is used to supervise their face existence prediction c^i to be close to 0 (i.e., no face).

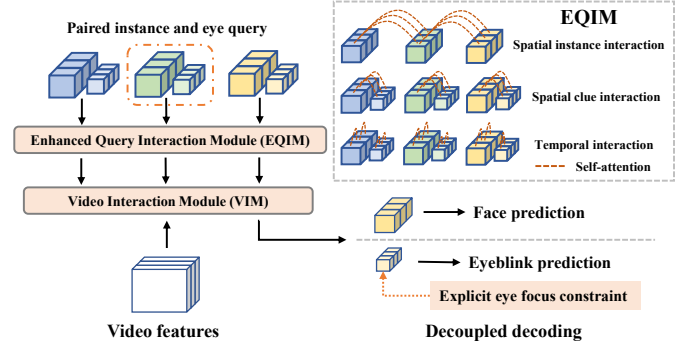


Fig. 14. Overview of the proposed InstBlink++.

V. INSTBLINK++

Despite the effectiveness, we have identified a potential limitation within InstBlink: due to the shared query design, the eyeblink feature may become highly entangled or fused with the global face features. While such a design can involve useful global face context for robust eyeblink representations, it lacks targeted modeling of the specific eye region features. Hence, we further introduce a concept of eye query, to explicitly capture fine-grained eye-specific features for enhancing discriminative power. This is further guaranteed by a decoupled decoding with eye focus constraint. We also build an additional interaction among eye query and instance query to enrich the eye query with global facial semantic context.

We call this new model as InstBlink++, which performs much more effectively while still retaining the real-time advantages of the one-stage inference property from InstBlink. An overview of the proposed InstBlink++ is shown in Fig. 14. The key enhancements lie in 3 folders: the introduction of paired eye query, the enhanced query interaction module, and the decoupled decoding with explicit eye focus constraint, which will be illustrated as follows.

A. Instance Query with Paired Eye Query

We introduce paired instance-eye queries to address feature coupling and better capture fine-grained eye-region information for effective eyeblink representation. While the concept of eye query also appears in MCGaze [55], it is designed for gaze modeling and lacks instance awareness. In contrast, our paired instance-eye query design explicitly encodes instance-level eye semantics and is tailored for fine-grained spatiotemporal eyeblink modeling. Specifically, each instance query Q^i in InstBlink++ is paired with one eye query q^i , which is of the same feature dimension with instance query (i.e., $q^i \in \mathbb{R}^{T \times D}$) but dedicated to model fine-grained instance-level eye semantics in the entire video clip. Similar with instance query, the eye query is also paired with a proposal tube $p^i \in \mathbb{R}^{T \times 4}$ (i.e., spatio-temporal bounding boxes across the time), which aims at indicating the eye location of i -th instance across the entire video clip. At the first iteration of each complete forward propagation, q^i and p^i are initialized by copying 2 learnable parameters $\bar{q}^i \in \mathbb{R}^{1 \times D}$ and $\bar{p}^i \in \mathbb{R}^{1 \times 4}$ T times along the temporal dimension. Ideally, under a proper learning strategy

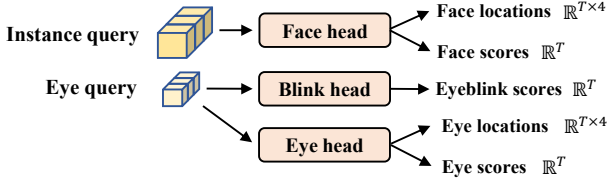


Fig. 15. Overview of the decoupled decoding.

(introduced later), the eye query will focus on attending to eye-region semantics, thereby enabling fine-grained spatiotemporal modeling of eye dynamics.

B. Enhanced Query Interaction Module (EQIM)

In addition to the instance-level feature interaction introduced in Sec. IV-B, to make the eye query better aware of the global face context for robust eye and eyeblink representation, we further introduce spatial clue interaction between the instance query and the paired eye query via self-attention:

$$\mathbf{Q}_t^i, \mathbf{q}_t^i = \text{MHSA}(\mathbf{Q}_t^i, \mathbf{q}_t^i), \quad i \in [1, N], t \in [1, T]. \quad (11)$$

Such an operation not only makes the eyeblink representation more robust due to the engagement of global face context, but also enriches the fine-grained representation within the instance query which boosts instance localization ability. Then, temporal modeling is also built within eye query via temporal feature interaction:

$$\{\mathbf{q}_t^i\}_{t=1}^T = \text{MHSA}(\{\mathbf{q}_t^i\}_{t=1}^T), i \in [1, N]. \quad (12)$$

Such temporal interaction enables fine-grained temporal modeling of eye dynamics, which is crucial for accurate eyeblink boundary localization, as verified in Sec. VI-F.

C. Decoupled Decoding with Eye Focus Constraint

The paired instance–eye queries pass through the video interaction module (VIM) to collect instance-specific and task-specific features (i.e., face-related or eye-related) from the video features. Unlike the initial InstBlink, the instance query here primarily captures global face features for face detection and tracking, while the eye query focuses on fine-grained eye features for eye-related tasks. This naturally achieves a form of decoupling, which is further enforced by a decoupled decoding strategy (Fig. 15): (1) The instance query is decoded solely by the face head in InstBlink for face identification and localization. (2) Different from InstBlink, we use blink head to decode features within eye query, rather than instance query, which is formulated as:

$$b^i = \text{Sigmoid}(MLP_b(\tilde{q}^i)), \quad (13)$$

where $b^i \in \mathbb{R}^T$ is the frame-level eyeblink scores and $\tilde{q}^i$ is the updated eye query feature after VIM. (3) We further introduce eye focus constraint to the eye query to explicitly let it focus on the fine-grained eye region. Specifically, we first apply an eye head on the eye query to equip it with the ability to capture the eye region. Given the updated eye query feature $\tilde{q}^i$, an

MLP layer with Sigmoid normalization is used to indicate the existence of human eyes as

$$c^i = \text{Sigmoid}(MLP_c(\tilde{q}^i)), \quad (14)$$

where $c^i \in \mathbb{R}^T$ denotes the eye classification scores across frames. Eye region localization is achieved similarly as:

$$l^i = MLP_l(\tilde{q}^i), \quad (15)$$

where $l^i \in \mathbb{R}^{T \times 4}$ denotes the eye bounding boxes across frames. It will also be used to renew the proposal tube p^i for eye-related feature extraction in VIM. During training, we define eye focus loss to let the eye query focus on the fine-grained local eye region:

$$\begin{aligned} \mathcal{L}_{eye}(\hat{y}^j, y^{\hat{\sigma}(j)}) = & \sum_{t=1}^T \left[\mathcal{L}_{cls}(\hat{c}_t^j, c_t^{\hat{\sigma}(j)}) \right. \\ & \left. + 1_{\{\hat{c}_t^j \neq 0\}} \left(\mathcal{L}_{box}(\hat{l}_t^j, l_t^{\hat{\sigma}(j)}) \right) \right], \end{aligned} \quad (16)$$

where $y^{\hat{\sigma}(j)}$ is the prediction that has been matched with the ground truth instance $\hat{y}^j$ according to the bipartite matching result. $\hat{c}^j \in \mathbb{R}^T$ and $\hat{l}^j \in \mathbb{R}^{T \times 4}$ indicate the frame-level eye existence and eye region position (i.e., bounding box). Overall, InstBlink++ is trained by minimizing the following objective over all matched ground-truth–prediction pairs:

$$\begin{aligned} \mathcal{L} = & \sum_{j=1}^{N_{gt}} \left[\mathcal{L}_{face}(\hat{y}^j, y^{\hat{\sigma}(j)}) + \mathcal{L}_{eye}(\hat{y}^j, y^{\hat{\sigma}(j)}) \right. \\ & \left. + \lambda \mathcal{L}_{blink}(\hat{y}^j, y^{\hat{\sigma}(j)}) \right]. \end{aligned} \quad (17)$$

The other training details of InstBlink++ are consistent with InstBlink unless otherwise stated.

VI. EXPERIMENTS

A. Experimental Settings

Dataset. Experiments are conducted on 3 datasets: (1) MPEblink 1.0*, which retains the same data as MPEblink 1.0 while incorporating improved annotation quality to enable a fairer and more reliable evaluation. (2) MPEblink 2.0, which is split into 540 training, 169 validation, and 212 test videos, ensuring that videos from the same movie appear in only one split. (3) HUST-LEBW [1], the only other unconstrained eyeblink dataset besides MPEblink. Although it is not designed for multi-person eyeblink detection, we include it to demonstrate the generalization ability of our method in single-person and trimmed cases.

Metrics. We report the metrics designed in Sec. III-F for MPEblink series. For HUST-LEBW, which focuses on single-person trimmed cases, we follow prior works and report Recall, Precision, and F1 score.

Implementation details. We use ResNet-50-FPN [79], [80] as our backbone network, which is pre-trained on YouTube-VIS [81] for a general instance representation ability. The query number N and iteration time M are set to 50 and 4 respectively. AdamW [82] optimizer with a batch size of 8 is used to train the model. The initial learning rate is set to 2.5e-5 for the backbone and 2.5e-4 for the other parts. At the training

TABLE II
THE MAIN RESULTS ON THE MPEBLINK 1.0* DATASET.

Method	IB	IB ₅₀	IB ₇₅	Blink-AP	Blink-AP ₅₀	Blink-AP ₇₅	Inst-AP	Inst-AP ₅₀	Inst-AP ₇₅
Soukupov’ a and Cech [13]	0.06	0.24	0.02	0.15	0.60	0.06	57.39	73.86	61.43
Blink detection+ [31]	0.07	0.29	0.04	0.18	0.71	0.09			
Daza et al. [6]	0.76	2.69	0.46	1.87	6.59	1.12			
Zeng et al. [5]	2.59	10.49	0.34	6.32	25.63	0.83			
BlinkFormer [28]	1.81	7.73	0.19	4.42	18.89	0.46			
InstBlink (Ours)	5.87	16.79	4.60	9.65	27.62	7.56	69.46	85.82	77.13
InstBlink++ (Ours)	7.69	23.37	5.38	12.48	37.91	8.73	73.98	90.47	84.16

TABLE III
THE MAIN RESULTS ON THE MPEBLINK 2.0 DATASET.

Method	IB	IB ₅₀	IB ₇₅	Blink-AP	Blink-AP ₅₀	Blink-AP ₇₅	Inst-AP	Inst-AP ₅₀	Inst-AP ₇₅
Soukupov’ a and Cech [13]	0.08	0.36	0.03	0.24	1.01	0.09	31.54	51.73	31.57
Blink detection+ [31]	0.06	0.26	0.02	0.17	0.73	0.06			
Daza et al. [6]	0.66	2.57	0.26	1.88	7.27	0.74			
Zeng et al. [5]	2.66	9.02	0.85	7.50	25.46	2.39			
BlinkFormer [28]	1.82	6.83	0.39	5.14	19.29	1.10			
InstBlink (Ours)	3.29	10.28	1.92	6.42	20.02	3.74	41.22	59.93	43.55
InstBlink++ (Ours)	4.67	13.67	2.97	8.58	25.13	5.46	52.54	73.60	59.01

stage, for memory efficiency, we sample frames at a half frame rate (~ 12 FPS), and the input clip length is set to 11 frames, which is longer than most eyeblink events (0.1–0.5 seconds, as shown in Fig. 11). Frames are resized to 640×360 before being fed into the network. Training runs for 10k iterations, with the learning rate reduced by 0.1 at iterations 6k and 9k.

During testing, the original frame rate is used with an input clip length of 36. We adopt sliding-window inference with a stride of 18 for long videos, and link predictions across adjacent clips based on face-box IoU. In this extension, we found that one cause of the instance localization failure stems from missed detections. This issue may partly arises from the limited number of queries due to our hardware limitations. Simply lowering the confidence threshold can partially alleviate this issue, but it also introduces redundant detections. Therefore, we combine this strategy with non-maximum suppression (NMS) [83]. Although this engineering modification does not improve Blink-AP, it leads to a substantial gain in Inst-AP. It is worth noting that the computational overhead of NMS is negligible in our setting, due to the sparse proposal property of the query-based framework.

Baselines. Multi-person eyeblink detection in untrimmed videos is a new task that has not been well concerned before. Therefore, we tailor the existing eyeblink detection approaches with a unified instance detection and tracking method. Specifically, we use the state-of-the-art face analysis toolbox Insight-Face [64] to achieve face [65] and landmark [63] detection. To track each instance, we link the face bounding boxes from the adjacent frames based on their similarity on box IoU. After obtaining the instance tracklets (i.e., a sequence of face bounding boxes and landmarks), we employ representative eyeblink detection methods [5], [6], [13], [28], [31] within each instance tracklets. Such a sequential pipeline is commonly used in the existing approaches under the single-person assumption.

B. Benchmarking Results on MPEblink Series

MPEblink 1.0*. The performance comparison on the MPEblink 1.0* dataset in Tab. II can be summarized as follows:

- All existing algorithms cannot achieve a satisfactory performance (i.e., $IB < 8\%$ and $IB_{50} < 25\%$) This indicates that eyeblink detection under multi-person, untrimmed, and unconstrained scenarios is indeed challenging and has not been well solved yet.
- InstBlink and InstBlink++ significantly outperform existing methods across all metrics. We attribute these gains primarily to three factors: (1) Our framework learns long-term eyeblink dynamics more effectively than frame-based [6] and sliding-window approaches [1], [5], [13], [28], [31]. (2) Effective face-eye feature utilization and joint optimization with global facial context (e.g., head pose, illumination) facilitates more robust blink representation, which sequential pipelines cannot exploit. This improved feature utilization also contributes to higher efficiency, as verified in Sec. VI-D. (3) The architecture also delivers higher Inst-AP by learning coherent spatio-temporal instance trajectories, whereas tracking-by-detection baselines lack sufficient modeling capacity and are more susceptible to occlusion.
- Compared with InstBlink, InstBlink++ achieves substantial performance gains, which fully demonstrates the effectiveness of the designed disentangled eye modeling.

MPEblink 2.0. The performance comparison on the MPEblink 2.0 dataset in Tab. III can be summarized as follows: (1) InstBlink++ still outperforms competing methods by a large margin, further demonstrating its strong generalization capability in both instance localization and fine-grained eyeblink modeling. (2) Compared with MPEblink 1.0*, existing methods exhibit consistently lower performance on MPEblink 2.0. This phenomenon highlights the increased difficulty and challenges introduced by the newly upgraded MPEblink 2.0 dataset. We further provide a quantitative analysis of difficulty attribution in Sec. VI-G.

TABLE IV

THE INFERENCE SPEED COMPARISON ON A SINGLE NVIDIA 3090 GPU, ASSUMING THE COMPARED METHODS USE INSIGHTFACE FOR FACE DETECTION (PER-FRAME TIME CONSUMPTION $T=9.3\text{ms}$, PER-FRAME COMPUTATIONAL COST GFLOPS $G=26.68$) AND LANDMARK DETECTION (5.4ms PER-FRAME, GFLOPS 6.1 PER-FRAME). #faces DENOTES THE FACE AMOUNT IN THE SCENE.

Metric/Method	InstBlink (Ours)	InstBlink++ (Ours)	Blink detection+ [31]	Daza et al. [6]	Zeng et al. [5]	BlinkFormer [28]
GFLOPs (36 frames)	36.7 (73.4 online)	39.3 (78.6 online)	$G (=960.5) + 221.0 \times \#faces$	$G + 221.4 \times \#faces$	$G + 551.5 \times \#faces$	$G + 236.0 \times \#faces$
Time/frame (ms)	11.5 (23.0 online)	13.5 (27.0 online)	$T (=9.3) + 5.4 \times \#faces$	$T + 9.1 \times \#faces$	$T + 26.5 \times \#faces$	$T + 10.8 \times \#faces$

TABLE V
MLLM EVALUATION ON MPEBLINK 2.0-S.

Method	Input Setting	Blink-AP	Blink-AP ₅₀
Qwen2.5-VL (zero-shot)	Full frame	Error	Error
Qwen2.5-VL (zero-shot)	Ground-truth face	0.06	0.28
Qwen2.5-VL (finetune)	Ground-truth face	2.43	8.72
Zeng et al [5]	Ground-truth face	6.51	25.97
InstBlink++	Full frame	10.00	35.16

TABLE VI
ACTION DETECTOR EVALUATION ON MPEBLINK 2.0.

Method	Input Setting	Blink-AP	Blink-AP ₅₀
Actionformer [84]	Ground-truth face	1.08	3.24
TriDet [85]	Ground-truth face	1.12	3.30
DyFADet [86]	Ground-truth face	1.14	2.87
DiGIT [87]	Ground-truth face	1.04	3.08
Zeng et al [5]	Ground-truth face	5.29	21.06
InstBlink++	Full frame	8.58	25.13

C. Evaluation on General Models

In this subsection, we investigate general models beyond blink-focused methods to uncover a broader picture of how these related approaches perform on the multi-person eyeblink detection task.

How do MLLMs perform on instance-aware eyeblink detection? While multimodal large language models (MLLMs) [18], [88]–[90] have demonstrated strong performance across various general multimodal tasks [91]–[96], their effectiveness on fine-grained eyeblink detection remains largely unexplored. In this work, we take a pioneering step to evaluate them on the fine-grained multi-person eyeblink detection task.

We select Qwen2.5-VL [18] for evaluation due to its strong video understanding and temporal awareness. Due to computational resource limitations, we test on videos shorter than 90 seconds (termed MPEblink 2.0-S), and evaluate its 3B version, which already demonstrates competitive performance on general benchmarks and is over 30 times larger than our model. From Tab. V, it can be observed that (1) In the practical full-frame setting, Qwen2.5-VL fails to generate valid outputs for joint multi-person localization and fine-grained eyeblink detection. (2) Even in a simplified setting, where ground-truth face regions are provided and the model only needs to detect eyeblink events, its performance remains unsatisfactory (far below that of the existing eyeblink-focused method [5] under the setting, and ours in the practical full-frame setting) even after fine-tuning. This highlights the challenges of instance-aware eyeblink detection in untrimmed videos and the limited capabilities of current MLLMs for such fine-grained tasks.

Can general action detectors handle fine-grained eyeblink

detection? We evaluate several representative methods [84]–[87] designed for general action detection (e.g., jumping) [70], [71]. The poor performance, as shown in Tab. VI, demonstrates that architectures designed for coarse-grained action detection cannot be effectively transferred to fine-grained eyeblink detection tasks. InstBlink++ outperforms these methods without any auxiliary input (e.g., face localization).

D. Inference Efficiency Analysis

The results are summarized in Tab. IV, assuming that all compared methods adopt InsightFace [64] for face and landmark detection, and that InstBlink (++) performs inference with a clip length of 36 frames. It can be observed that the InstBlink series achieves not only high inference speed but also significantly lower computational complexity. The core reasons lie in the inherent advantages of the one-stage network design adopted by InstBlink and InstBlink++:

(1) Effective feature utilization. The one-stage design enables extensive feature sharing across subtasks (e.g., instance localization, eyeblink modeling), unlike multi-stage pipelines where features cannot be effectively reused. As shown in Tab. IV, a significant portion of runtime and GFLOPs in multi-stage methods is consumed by face and landmark detection. Additionally, existing SOTA methods, trained on trimmed clips, rely on dense sliding-window inference for untrimmed videos, introducing computational redundancy. In contrast, our one-stage design performs joint modeling in a single pass with high feature utilization.

(2) Flexible offline and online inference. The InstBlink series is versatile for both offline and online inference. In offline scenarios, it processes entire video clips in a single forward pass to produce instance-level predictions. For longer videos or streaming, it operates in a near-online fashion using sliding window inference with partial overlap. The results within the overlap are averaged to mitigate prediction disruptions. This introduces a slight speed reduction (less than 15 ms extra per frame) while maintaining real-time performance ($\text{FPS} > 35$). Omitting the overlap leads to an acceptable accuracy drop, with less than 3% degradation on Blink-AP₅₀ and 1% on Blink-AP₇₅.

(3) $O(1)$ complexity. Thanks to the one-stage formulation, InstBlink and InstBlink++ maintain constant inference complexity with respect to the number of faces. In contrast to multi-stage methods, whose runtime and GFLOPs scale linearly with the instance count, our approach performs joint modeling in a single forward pass, rendering it largely insensitive to the number of people present in the scene.

Moreover, compared with InstBlink, the extended InstBlink++ achieves a substantial performance improvement (as demonstrated in Tab. II and Tab. III) while introducing only a

TABLE VII
PERFORMANCE ON THE HUST-LEBW DATASET.

Type	Method	Eye	Recall	Precision	F1
Multi-stage	Soukupová and Cech [13]	Left	36.07	64.71	46.32
		Right	30.16	57.58	39.58
	Hu et al. [1]	Left	54.10	89.19	67.35
		Right	44.44	76.71	56.28
	Daza et al. [8]	Left	96.03	60.80	74.46
		Right	79.50	73.48	76.37
	Zeng et al. [5]	Left	91.80	89.60	90.69
		Right	91.27	92.74	92.00
	Blink detection.+ [31]	Both	58.99	80.05	67.90
		Both	93.39	75.33	83.39
One-stage	Daza et al. [6]	Both	86.69	82.06	84.31
	BlinkFormer [28]	Both	85.80	85.10	85.30
	DE-ViT [29]	Both	91.34	76.82	83.45
One-stage	InstBlink	Both	92.13	85.40	88.63
	InstBlink++	Both			

TABLE VIII
EFFECT OF PAIRED INSTANCE-EYE QUERY.

Method	IB	Blink-AP	Blink-AP ₅₀	Inst-AP
w/o paired eye query	3.50	6.35	20.15	51.88
Full model	4.67	8.58	25.13	52.54

marginal increase in inference overhead, and it still preserves the inherent advantages of the one-stage design discussed above. This further highlights the superiority and practical viability of the InstBlink++ architecture.

E. Main Result on HUST-LEBW

We also evaluate InstBlink on the HUST-LEBW dataset [1] to assess its generalizability in single-person, trimmed, in-the-wild scenarios. Due to HUST-LEBW’s limitations (e.g., small scale, lack of instance-level localization annotations, and missing frame-level eyeblink labels), it is not suitable for training MPEblink to jointly learn instance and temporal eyeblink localization. Instead, we use a cross-dataset evaluation protocol (trained on MPEblink but directly test on HUST-LEBW), which presents a more challenging and practical setting, following prior practice [19].

As shown in Tab. VII, InstBlink++ achieves competitive performance with the state-of-the-art, without any dataset-specific fine-tuning, demonstrating its strong generalization ability. Unlike multi-stage methods that rely on pre-located facial regions, InstBlink++ performs face detection, tracking, and eyeblink detection in a single stage with higher efficiency (as shown in Sec. VI-D), without the need for auxiliary inputs or models. This design offers greater versatility and broader applicability in real-world scenarios.

F. Ablation Studies

We perform ablation studies on MPEblink 2.0 to evaluate the contributions of each model component in InstBlink++.

Paired instance-eye query. From Tab. VIII, it can be observed that introducing the eye query leads to a significant performance improvement, increasing Blink-AP from 6.35 to 8.58 (a 35% relative gain), which verifies the essential role of the eye query in fine-grained eyeblink modeling.

TABLE IX
ABLATION STUDIES ON EQIM.

Method	IB	Blink-AP	Blink-AP ₅₀	Inst-AP
w/o temporal interaction	0.90	1.67	6.54	48.00
w/o spatial interaction	3.16	6.21	20.39	46.44
w/o clue interaction	4.47	8.26	25.51	51.62
Full model	4.67	8.58	25.13	52.54

TABLE X
EFFECT OF QUERY-AWARE FILTER OPERATION.

Method	IB	Blink-AP	Blink-AP ₅₀	Inst-AP
w/o filter operation	3.42	6.33	20.72	50.49
Full model	4.67	8.58	25.13	52.54

Temporal modeling in EQIM. Here we examine the role of temporal modeling achieved through intra-query feature interactions along the temporal dimension in the EQIM module (Sec. V-B). As shown in the second row of Tab. IX, removing temporal interaction leads to a drastic drop in Blink-AP (from 8.58% to 1.67%), demonstrating that temporal modeling is critical for eyeblink detection in untrimmed video (i.e., only using appearance feature cannot accurately localize eyeblink). It can also be observed that the temporal interaction facilitates the performance of Inst-AP. We argue that this is because it can also strengthen instance-level association among the features across frames to facilitate the instance tracking ability.

Spatial modeling in EQIM. From the last three rows of Tab. IX, we observe that: (1) Spatial interaction among instance queries can boost both Blink-AP and Inst-AP by large margins, as it can build strong communication among queries to better model the instance features under complex circumstances such as occlusion due to human interactions. (2) Spatial clue interaction between paired instance and eye query improve the performance by a noticeable margin, as this operation can lead the eye query to be better aware of the global face context for robust eyeblink representation. On the other hand, the instance query can also benefit from the eye query by enriching its fine-grained feature representation capability, leading to improved Inst-AP.

Query-aware filter operation in VIM. As in Tab. X, compared with directly using RoI features for query update, using query-aware filtered RoI features can boost both Blink-AP and Inst-AP by large margins. We speculate that the filter operation can activate task-specific RoI features and resist background. Thus, finer face and eyeblink clues can be collected to update the query.

Decoupled decoding. From Table XI, we can observe that removing the decoupled decoding (here means the instance query will take responsibility for both face and eyeblink decoding) leads to a significant performance drop on both Blink-AP and Inst-AP. This essentially demonstrates the effectiveness of the proposed decoupled decoding, as it can solve the feature entangle problem and also switch the eyeblink decoding from global face feature into local eye feature, which enables a more fine-grained eyeblink prediction ability.

Multi-task learning mechanism. Here we study the effective-

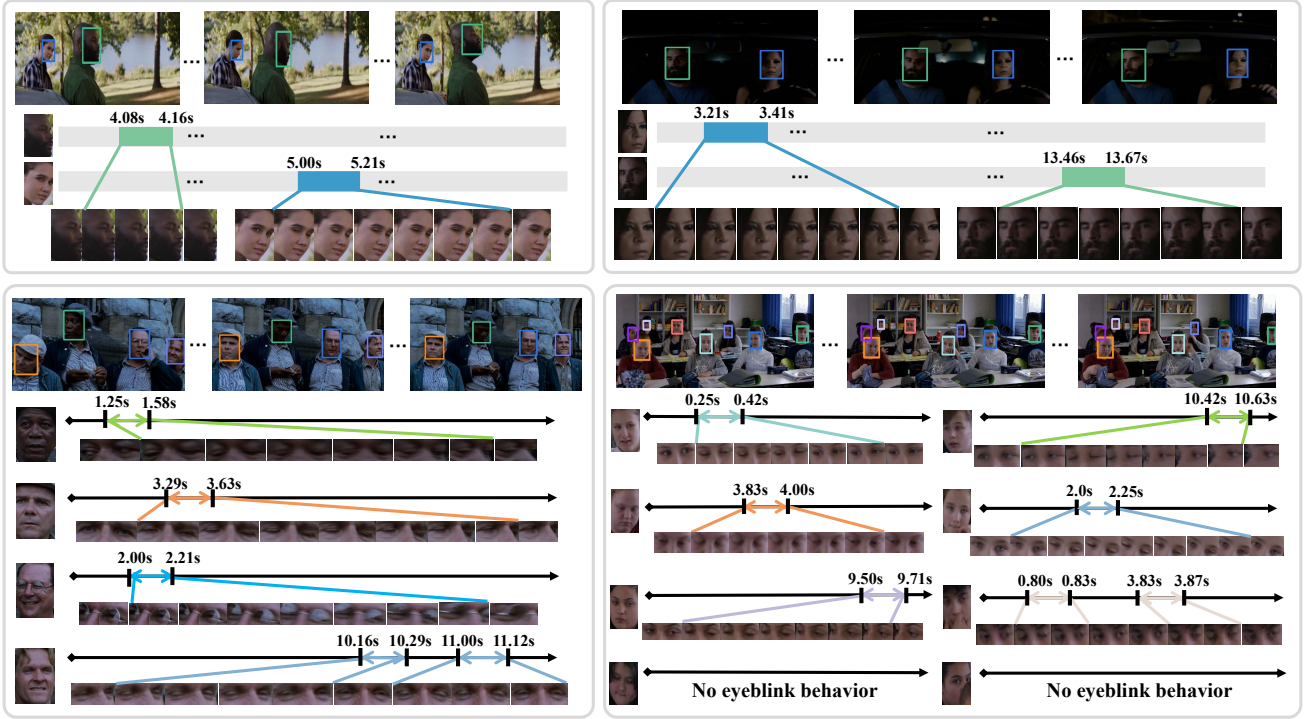


Fig. 16. Qualitative analysis for successful cases.

TABLE XI
EFFECT OF DECOUPLED DECODING.

Method	IB	Blink-AP	Blink-AP ₅₀	Inst-AP
w/o decoupled decoding	3.90	7.12	21.61	49.89
Full model	4.67	8.58	25.13	52.54

TABLE XII
STUDIES ON MULTI-TASK LEARNING MECHANISM.

Method	IB	Blink-AP	Blink-AP ₅₀	Inst-AP
w/o eye focus constraint	4.11	7.51	22.42	52.99
w/o blink detection	0.00	0.00	0.00	51.63
Full model	4.67	8.58	25.13	52.54

ness of our framework from a multi-task learning perspective. As shown in Tab. XII, eyeblink detection accuracy improves substantially when the explicit eye-focus constraint is imposed on eye queries, as this objective explicitly guides eye queries to successfully learn to focus on eye-region semantics and model fine-grained eye dynamics, with the ultimate goal of effective eyeblink detection. In addition, learning eyeblink detection also has a positive effect on Inst-AP, indicating that face detection and tracking benefit from eye-related cues.

G. Difficulty Attribution

Here, we provide a fine-grained, quantitative error analysis to identify the real challenging factors that still hinder model performance. We then present qualitative results for grounded and illustrative understanding in Sec. VI-H.

We categorize the MPEblink 2.0 test set into four attribute subsets: (1) Extreme pose: Samples with extreme head pose and gaze deviations, within the top 10% of their respective

TABLE XIII
DIFFICULTY ATTRIBUTION. HIGHLIGHTED ENTRIES INDICATE METRICS LOWER THAN THE FULL (BASELINE) SET.

Attribute Subset	IB	Blink-AP	Blink-AP ₅₀	Inst-AP
Full (baseline)	4.67	8.58	25.13	52.41
Extreme pose	1.62	2.82	8.99	67.54
Occlusion	4.56	9.21	30.08	35.91
Visual discontinuity	6.82	17.04	49.28	10.88
Small faces	1.81	3.33	10.17	48.72

distributions, as estimated by off-the-shelf estimators [55], [69]. (2) Occlusion: Samples with at least partial occlusion of the face. (3) Visual discontinuity: Samples affected by visual discontinuities, such as face invisibility from character movements or abrupt face location changes due to shot transitions. (4) Small faces: Samples where the average area of the ground-truth face bounding box is less than 1% of the total video resolution.

From Tab. XIII, we observe the following findings: (1) Extreme head pose results in a significant drop in eyeblink detection performance (low Blink-AP). (2) Occlusion and visual discontinuity severely impact instance tracking, as reflected in the low Inst-AP scores, with visual discontinuities causing the most disruption. This indicates the model's limited ability for long-term instance identification independent of short-term appearance cues, highlighting the need for future architectural improvements. Note that the seemingly inflated Blink-AP scores in these cases are due to the evaluation protocol, as Blink-AP is computed only over true positive tracking instances, which may be less challenging. (3) Small face scales significantly degrade both fine-grained blink detection and relatively coarse-grained face tracking accuracy.

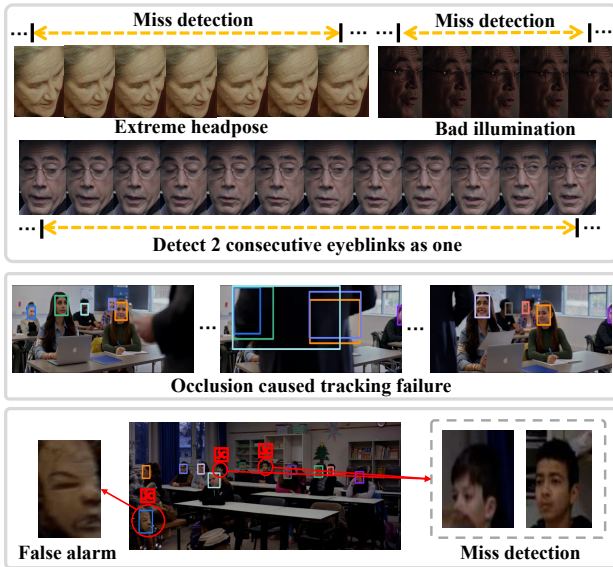


Fig. 17. Qualitative analysis for failure cases.

H. Qualitative Analysis

Here we give some qualitative results to study the success and failure cases of the proposed InstBlink++. From Fig. 16, we can see that the proposed method can aware the attendees robustly and detect their eyeblinks under various circumstances, including different head poses, lighting conditions, lens reflections, and relatively crowded scenes.

Nevertheless, it can also be observed from Fig. 17 that our method will also fail to work in some cases. Firstly, miss detection in eyeblinks may occur under extreme head pose or illumination conditions. Secondly, sometimes it cannot distinguish the rapid consecutive eyeblinks. Such a phenomenon actually highlights the essential challenge of detecting eyeblinks in untrimmed videos. Moreover, our methods can also fail in the aspect of instance localization and tracking, under the factors including occlusion and small faces in crowd scenes.

VII. CONCLUSIONS

In this work, we shed the light on a new research task termed multi-person eyeblink detection in the wild for untrimmed video. We formally define the task and introduce a tailored dataset MPEblink, with interpretable metrics for proper model evaluation. To perform multi-person eyeblink detection in untrimmed videos, we propose the first one-stage multi-person eyeblink detection framework called Instblink, along with its improved version, Instblink++. Extensive experimental results demonstrate the superiority of InstBlink and Instblink++ in both effectiveness and efficiency, while also revealing the unsatisfactory performance even in SOTA methods (i.e., IB < 5%). This verifies the challenges of unconstrained multi-person eyeblink detection in untrimmed long videos. We hope our work can pave the way for eyeblink detection in more general settings and motivate future research in relevant fields.

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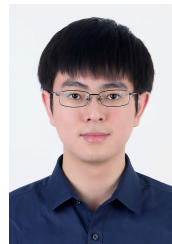
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